SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA16   COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)
Assessment Tool Weightage: 10%

QUESTIONS: 30   **Total Marks:30**

Annexure A

MCQ TEST

Criteria	Conceptual Understanding ***(3)***	Accuracy of Answers ***(2.5)***	Application of Knowledge ***(2)***	Time Management ***(1.5)***	Question Interpretation ***(1)***	***Total(10)***
Weightage	*30%*	25%	20%	15%	10%	*100%*
Q						

Code of Conduct:

I Bharath P / 192511282 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Bharath

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANNEXURE A

MCQ Test:

1.Consider a simple linear regression model with one independent variable. Which of the following is true? (CO3)
Option_a: The regression line always passes through the mean of X and Y.
Option_b: The slope is always positive.
Option_c: The intercept is always positive.
Option_d: The correlation coefficient is always 1.

2.Which of the following statements is correct about Decision Trees? (CO3)
Option_a: Decision trees can only handle numerical data.
Option_b: Decision trees are always prone to overfitting.
Option_c: Decision trees can be used for classification and regression.
Option_d: None of the above.

3.Which of the following methods can be used for pruning in decision trees? (CO3)
Option_a: Pre-pruning
Option_b: Post-pruning
Option_c: Both (a) and (b)
Option_d: None of the above.

4.Which of the following is a disadvantage of Decision Trees? (CO3)
Option_a: Prone to overfitting.
Option_b: Requires large datasets.
Option_c: Cannot handle categorical data.
Option_d: Always has a high accuracy.

CSA16 – Data Warehousing and Data Mining
Assessment Tool 3 – MCQ Questions with Answers

Name: Bharath P

Reg No: 192511282

Signature: Bharath

1. Consider a simple linear regression model with one independent variable. Which of the following is true?

a) The regression line always passes through the mean of X and Y.

b) The slope is always positive.

c) The intercept is always positive.

d) The correlation coefficient is always 1.

Answer: a) The regression line always passes through the mean of X and Y.

2. Which of the following statements is correct about Decision Trees?

a) Decision trees can only handle numerical data.

b) Decision trees are always prone to overfitting.

c) Decision trees can be used for classification and regression.

d) None of the above.

Answer: c) Decision trees can be used for classification and regression.

3. Which of the following methods can be used for pruning in decision trees?

a) Pre-pruning

b) Post-pruning

c) Both (a) and (b)

d) None of the above.

Answer: c) Both (a) and (b)

4. Which of the following is a disadvantage of Decision Trees?

a) Prone to overfitting.

b) Requires large datasets.

c) Cannot handle categorical data.

d) Always has a high accuracy.

Answer: a) Prone to overfitting.

5. What does Bayes' Theorem primarily help with in probability and statistics?

a) Determining prior probabilities only

b) Updating probabilities based on new evidence

c) Computing standard deviation of a dataset

d) Finding mean and median in distributions

Answer: b) Updating probabilities based on new evidence

6. In Bayes' Theorem, what does the term P(A|B) represent?

a) The probability of event B given event A has occurred

b) The probability of event A given event B has occurred

c) The joint probability of A and B

d) The marginal probability of B

Answer: b) The probability of event A given event B has occurred

7. A factory has two machines, A and B. Machine A produces 60% of the products, and machine B produces 40%. If 2% of A's products and 3% of B's products are defective, what is the probability that a randomly selected defective product came from Machine B?

a) 0.40

b) 0.50

c) 0.60

d) 0.75

Answer: b) 0.50

8. Which of the following is not required to apply Bayes' Theorem?

a) Prior probability of the event

b) Conditional probability of the evidence

c) Total number of possible outcomes

d) Probability of the evidence

Answer: c) Total number of possible outcomes

9. A doctor uses a test to diagnose a disease that affects 1% of a population. The test correctly identifies the disease 95% of the time but also gives a false positive in 5% of cases. If a patient tests positive, what is the probability they actually have the disease?

a) 0.16

b) 0.34

c) 0.95

d) 0.50

Answer: a) 0.16

10. Bayes' Theorem is most useful when dealing with which type of probability?

a) Independent events

b) Conditional probability

c) Mutually exclusive events

d) Uniform probability distribution

Answer: b) Conditional probability

11. If a hypothesis H has a prior probability of 0.3 and new evidence E has a likelihood of 0.8 given H, how does Bayes' Theorem adjust the probability of H after seeing E?

a) The probability of H increases

b) The probability of H decreases

c) The probability of H remains unchanged

d) The probability of H is completely unknown

Answer: a) The probability of H increases

12. A bag contains 5 red and 7 blue balls. A ball is drawn randomly and found to be red. What is the probability that the next ball drawn is also red (assuming the first ball is not replaced)?

a) 5/12

b) 4/11

c) 5/11

d) 6/12

Answer: b) 4/11

13. Which field heavily relies on Bayes' Theorem for updating beliefs based on new data?

a) Machine Learning

b) Geometry

c) Classical Physics

d) Cryptography

Answer: a) Machine Learning

14. Why is backpropagation crucial in training deep neural networks?

a) It prevents overfitting by pruning unnecessary weights.

b) It updates weights using the gradient of the loss function.

c) It ensures each layer has the same activation values.

d) It removes redundant layers to optimize the model.

Answer: b) It updates weights using the gradient of the loss function.

15. During backpropagation, what happens when the learning rate is too high?

a) The model converges faster with high accuracy.

b) The model may overshoot the optimal weight values and fail to converge.

c) The model will always get stuck in local minima.

d) The number of hidden layers increases automatically.

Answer: b) The model may overshoot the optimal weight values and fail to converge.

16. What is the main advantage of rule-based induction models?

a) They provide an interpretable set of rules for decision-making.

b) They work well with both numerical and categorical data.

c) They can generalize well without requiring large datasets.

d) Both (a) and (b)

Answer: a) They provide an interpretable set of rules for decision-making.

17. Which of the following is an example of a rule-based induction algorithm?

a) Decision Tree

b) Naïve Bayes

c) Support Vector Machine

d) K-Nearest Neighbors

Answer: a) Decision Tree

18. In rule-based induction, what happens when the rules are too specific to the training data?

a) The model may suffer from overfitting.

b) The model will generalize better.

c) The rules will be shorter and more efficient.

d) The accuracy will always increase.

Answer: a) The model may suffer from overfitting.

19. Which approach is commonly used to simplify rule-based induction models?

a) Pruning redundant or less significant rules

b) Adding more features to the dataset

c) Increasing the depth of the decision tree

d) Using more training samples

Answer: a) Pruning redundant or less significant rules

20. What is the primary purpose of a confusion matrix in evaluating a classifier?

a) To visualize the performance of a classifier by showing correct and incorrect predictions

b) To calculate the computational complexity of the classifier

c) To determine the training time of the classifier

d) To identify the features used by the classifier

Answer: a) To visualize the performance of a classifier by showing correct and incorrect predictions

21. Which metric is used to measure the proportion of correctly classified instances out of the total instances?

a) Precision

b) Recall

c) Accuracy

d) F1-Score

Answer: c) Accuracy

22. In a binary classification problem, what does the True Positive (TP) value represent?

a) The number of negative instances correctly classified as negative

b) The number of positive instances correctly classified as positive

c) The number of positive instances incorrectly classified as negative

d) The number of negative instances incorrectly classified as positive

Answer: b) The number of positive instances correctly classified as positive

23. What is the formula for calculating precision in a classification problem?

a) Precision = TP / (TP + FP)

b) Precision = TP / (TP + FN)

c) Precision = (TP + TN) / (TP + TN + FP + FN)

d) Precision = TP / (FP + FN)

Answer: a) Precision = TP / (TP + FP)

24. Which of the following metrics is most useful when the classes are imbalanced?

a) Accuracy

b) F1-Score

c) Recall

d) Specificity

Answer: b) F1-Score

25. What does a high recall value indicate in a classification model?

a) The model has a low number of false positives

b) The model has a low number of false negatives

c) The model has a high number of true negatives

d) The model has a high number of false positives

Answer: b) The model has a low number of false negatives

26. Which of the following is an example of a rule-based induction algorithm?

a) Decision Tree

b) Naïve Bayes

c) Support Vector Machine

d) K-Nearest Neighbor

Answer: a) Decision Tree

27. In rule-based induction, what happens when the rules are too specific to the training data?

a) The model may suffer from overfitting.

b) The model will generalize better.

c) The rules will be shorter and more efficient.

d) The accuracy will always increase.

Answer: a) The model may suffer from overfitting.

28. Which approach is commonly used to simplify rule-based induction models?

a) Pruning redundant or less significant rules

b) Adding more features to the dataset

c) Increasing the depth of the decision tree

d) Using more training samples

Answer: a) Pruning redundant or less significant rules

29. What is the primary purpose of a confusion matrix in evaluating a classifier?

a) To visualize the performance of a classifier by showing correct and incorrect predictions

b) To calculate the computational complexity of the classifier

c) To determine the training time of the classifier

d) To identify the features used by the classifier

Answer: a) To visualize the performance of a classifier by showing correct and incorrect predictions

30. Which metric is used to measure the proportion of correctly classified instances out of the total instances?

a) Precision

b) Recall

c) Accuracy

d) F1-Score

Answer: c) Accuracy